

Project Team Registration

Please note that each project team must consist of 5–6 members.

Team Registration Form (for teams that are already formed):

Link: <https://forms.gle/cjnRL6QSg4o2Zud87>

Submission Deadline: Tuesday, March 17, 2026 – 8:00 PM

Individual Registration Form (for students who have not yet joined a team):

Link: <https://forms.gle/hZLxSgzvi9Mk3CFz7>

Form opens: Tuesday, March 17, 2026 – 8:00 PM

Form Closes: Thursday, March 19, 2026 – 8:00 PM

Medical Diagnosis Expert System

Description:

Develop an expert system that can diagnose common diseases based on symptoms provided by the user. The system should utilize an appropriate knowledge representation method to link diseases with their symptoms. To implement this method, suitable libraries such as Experta, networkx, Kanren, or others can be used. Natural Language Processing (NLP) will process user input in free text, converting raw symptom descriptions into structured data. The system should be capable of handling tasks like tokenization, lemmatization, and stemming to accurately interpret and understand symptoms from natural language queries. Once a disease is diagnosed, the system will also display relevant precautions or recommendations to the user based on the identified condition.

Datasets:

The dataset consists of three columns:

1. **Disease:** The name of the disease.
2. **Symptoms:** A list of symptoms associated with each disease, separated by commas.
3. **Precautions:** A set of precautions or recommendations related to each disease that can be displayed to users.

Deliverables

The developed system should include the following features:

The application should behave similarly to a chatbot or expert consultation system, where the user interacts with the system through a conversational interface. The user provides symptoms in natural language, and the system analyzes them to suggest possible diseases based on the knowledge base.

The system should provide a simple and user-friendly interface where users can:

- Enter their symptoms in text form.
- View the system responses and questions.
- See the diagnosis results and recommendations.

If the system cannot determine the disease with enough certainty, it should continue asking the user additional questions about other possible symptoms in order to narrow down the diagnosis and reach a more accurate conclusion.

If multiple diseases match the provided symptoms, the system should display a percentage of certainty (confidence level) for each possible disease.

Example:

- Flu – 70% certainty
- Common Cold – 20% certainty
- Allergy – 10% certainty

After determining the most probable disease, the system should display precautions or health recommendations from the dataset to help the user manage the condition or reduce symptoms.

Run Example

Kindly note that this is just a simple example to help you envision how the input and output would work. The input and output shown here may not exactly fit your dataset, and the actual behavior of the system will depend on how you handle the data and the way you implement the system.

User Input:

"I have itching and skin rash."

System Output:

- **Diagnosis:** Based on the symptoms you've provided, it seems you might have Fungal Infection.
- **Precautions:**
 - Bath twice daily.
 - Use Dettol or neem in your bathing water.
 - Keep the infected area dry.
 - Use clean clothes to avoid further irritation or infection.